Kwanho Kim

C++ Software Engineer

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SUMMARY

Software engineer focused on C++ development, error recovery and software integration. Experience at KLA, Pickit 3D and Bosch spans equipment, robotics and vehicle communication.

EXPERIENCE

KLA — C++ Engineer

Aug 2026–Present · Zeta team, ICOS Division · Leuven, Belgium

- Analyse lifter fault recovery and design changes across application and controller software; early implementation is in progress with team review.
- Work on camera white-balance settings, device readback and the values shown in the interface.
- Refactor serial device-control integration and logging within a shared equipment layer.

Pickit 3D — Research & Software Engineer

Aug 2025–Mar 2026 · Leuven, Belgium

- Worked on system design and integration for robotic-vision software, focusing on communication interfaces and failure handling.
- Designed a Python WebSocket client with reconnect, backoff and message validation for internal integrations.
- Traced a field communication failure to a port conflict and implemented the fix.

Bosch — V2X Communication Intern

Jun 2024–Nov 2024 · Hildesheim, Germany

- Built a vehicle-to-everything (V2X) demonstrator, defined message interfaces and checked end-to-end behaviour in test runs.
- Integrated communication modules and developed a React interface for an eBike safety prototype.

SKILLS

Programming: C++, C, Python, TypeScript, JavaScript.

Systems & integration: Error handling and recovery, device communication, WebSocket, REST, ROS 1/2.

Applications & tools: React; Git, Docker, Linux.

Spoken languages: Korean (native); English (IELTS Academic 7.5, Mar 2025).

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SELECTED PROJECTS

Color Saves Life — Bosch ConnectedExperience 2024

Feb 2024 · 1st place, digital.auto & mobility challenge · Berlin

- Led and presented a hackathon team; integrated eye-tracking with a simulation pipeline and built a plugin-based Python client.

Autonomous Lane-Keeping Assist — SEA:ME

Aug–Dec 2023 · Wolfsburg, Germany

- Implemented CNN-based lane detection in CARLA and a client to collect and evaluate simulation data.

EDUCATION

Bachelor's, Mechatronics (in progress) — KNOU Prime College
Feb 2026–Present · Korea National Open University

Bachelor's, Artificial Intelligence — Korea National Open University
Mar 2023–Feb 2025 · Awarded

Bachelor's, Business Administration — Academic Credit Bank System
Apr 2019–Aug 2022 · Awarded

Bachelor's, Computer Engineering — Academic Credit Bank System
Apr 2019–Aug 2020 · Awarded

TRAINING & CERTIFICATION

SEA:ME — Automotive & Mobility program
Jul 2023–Jun 2024 · Completed · Wolfsburg, Germany

42 — Software engineering program
Completed · Seoul / Wolfsburg

Engineer Information Processing
Nov 2020 · Human Resources Development Service of Korea (HRD Korea)

LEADERSHIP

Rush03 — Keyboardist & leader
Aug 2022–Jul 2023 · 42 student club · Seoul, South Korea

- Led a 50-member community and delivered three events for 200–300 attendees.